TOP-Q — Day 8 worked notes

Mach–Zehnder interferometers, meshes, and how errors add up

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Labels as before. Everything here uses the *same* sign convention as Days 1–6 ($d\boldsymbol{a}/dz = iH\boldsymbol{a}$), so the array results can be plugged in on Days 9–10. Simulations: 5 random target meshes × 1000 error samples per point. (Day 7 of the original plan was the buffer; numbering is kept.)

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1 Building blocks

1.1 Directional coupler = a two-waveguide array [Derived; Verified-by-test]

Two identical guides coupled over a length ℓ: $d\boldsymbol{a}/dz = i\kappa\sigma_x\boldsymbol{a}$, so

$$B = e^{i\kappa\ell\sigma_x} = \begin{pmatrix} \cos\kappa\ell & i\sin\kappa\ell \\ i\sin\kappa\ell & \cos\kappa\ell \end{pmatrix} = \begin{pmatrix} \sqrt{\eta} & i\sqrt{1-\eta} \\ i\sqrt{1-\eta} & \sqrt{\eta} \end{pmatrix}, \qquad \eta = \cos^2\kappa\ell.$$

η = power that goes straight through ("bar"). 50:50 needs $\kappa\ell = \pi/4$. The test checks this equals the Day-5 propagator for a 2-guide array. **Link to Day 3** [Derived]: a gap error changes κ by a fraction ϵ_κ; at 50:50, $d\eta = -\sin(2\kappa\ell)\,\kappa\ell\,\epsilon_\kappa = -\frac{\pi}{4}\epsilon_\kappa$. So **coupler splitting error** $\sigma_\eta = \frac{\pi}{4}\sigma_\kappa$ — the same gap errors as on Day 3.

1.2 Phase shifter

$P(\phi) = \mathrm{diag}(e^{i\phi}, 1)$ on the top arm. On a chip this is usually a small heater: heating changes the effective index by Δn over a length L_h, giving $\phi = \frac{2\pi}{\lambda}\Delta n\, L_h$. **Link to Days 4–6**: an unwanted propagation-constant error $\delta\beta$ in an arm of length L is a phase error $\delta\phi = \delta\beta L$.

1.3 The MZI [Derived; Verified-by-test]

$T(\theta, \phi) = B\,P(\theta)\,B\,P(\phi)$. Multiplying out the middle (ideal couplers):

$$B\,P(\theta)\,B = i\,e^{i\theta/2}\begin{pmatrix} \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \\ \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \end{pmatrix} \;\Rightarrow\; |T_{11}|^2 = \sin^2\tfrac{\theta}{2}\ \text{(bar)}, \quad |T_{21}|^2 = \cos^2\tfrac{\theta}{2}\ \text{(cross)}.$$

θ sets the *split* (any ratio from 0 to 1); ϕ sets the *relative phase* at the input. Every factor is unitary, so T is unitary: no light is lost or created [Verified-by-test, $\|T^\dagger T - \mathbb{1}\| < 10^{-12}$].
In simple words: an MZI is a programmable beam splitter: one knob decides how the light divides, the other decides its phase.

1.4 Imperfect couplers limit the extinction [Derived; Verified-by-test]

With both couplers at $\eta \neq \frac{1}{2}$: $|T_{21}|^2 = 2\eta(1-\eta)(1+\cos\theta)$, so the **bar state is still perfect** ($\theta = \pi$ gives 0 cross), but the **cross state is not**: at most $4\eta(1-\eta)$. Simulated: 0.99 for $\eta = 0.45$, 0.96 for $\eta = 0.40$ (Fig. 2a). Correcting such imperfections by design is discussed by D. A. B. Miller, "Perfect optics with imperfect components", *Optica* (2015) [Literature — verify volume/page before citing].

2 Meshes

Any $N \times N$ unitary can be built from $N(N-1)/2$ MZIs plus N output phases. Reck *et al.* (1994) used a triangle; Clements *et al.*, *Optica* **3**, 1460 (2016), doi:10.1364/OPTICA.3.001460, showed a **rectangular** layout with the same count but only N columns and equal path lengths, which is more tolerant of loss [Literature (verified bibliographic record)]. We use the rectangular layout (Fig. 1). **Choice (stated):** we do not decompose a given target matrix; we draw random settings $\theta, \phi \in [0, 2\pi)$ and treat the resulting clean mesh U_0 as the target. For small errors the results below do not depend on the settings.

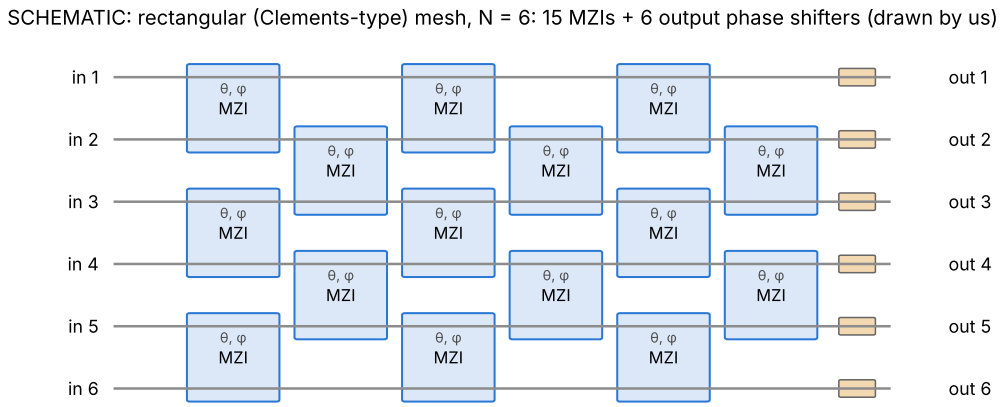


Figure 1: **Schematic drawn by us**, generated from `clements_layout(6)`: 15 MZIs in 6 columns, then 6 output phase shifters.

2.1 How good is a mesh? Two numbers

Normalised fidelity

$$F = \frac{|\mathrm{Tr}(U_0^\dagger U)|^2}{N\,\mathrm{Tr}(U^\dagger U)}.$$

$F = 1$ only if U equals U_0 up to a global phase and a uniform loss; $F \leq 1$ always (Cauchy–Schwarz) [Verified-by-test]. **Output-intensity error**: for each single-port input, the fraction of light that ends up in the wrong output ports (total-variation distance), averaged over inputs. 0 = perfect, 1 = completely wrong.

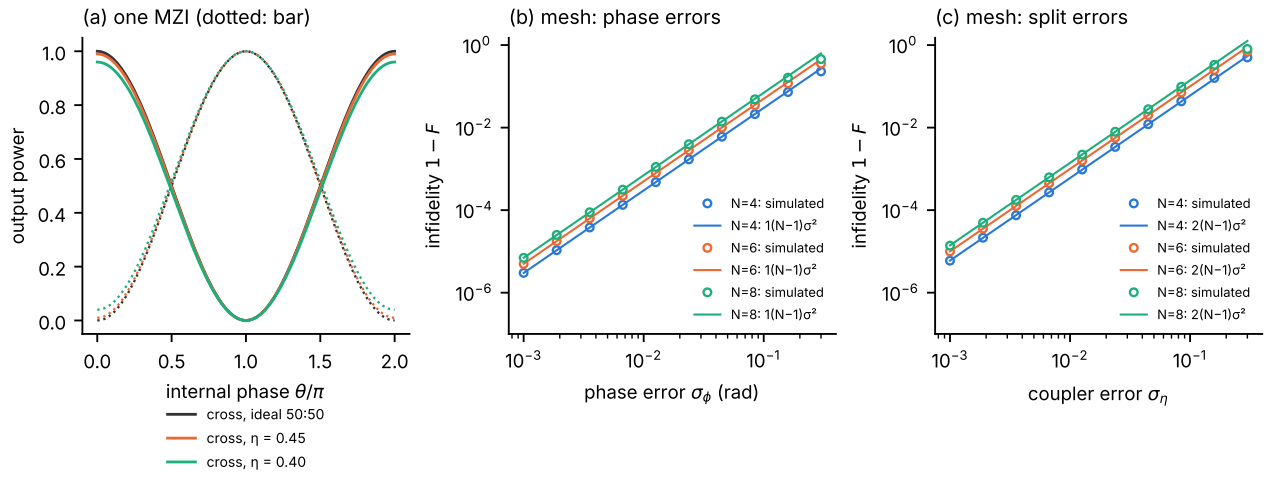


Figure 2: **Simulation**, `scripts/day8_mzi_mesh.py` → `data/day8_mzi_mesh.npz`. (a) One MZI: cross (solid) and bar (dotted) power for ideal and imperfect couplers. (b,c) Mesh infidelity vs. error size: circles = simulation, lines = first-order formulas.

3 How errors add up [Derived; Simulated]

3.1 Phase errors

One phase error δ on one mode acts like $W = e^{i\delta P}$ (P = projector on that mode inside the mesh). Then $\mathrm{Tr}\, W = N-1+e^{i\delta}$ and $|\mathrm{Tr}W|^2/N^2 \approx 1-(N-1)\delta^2/N^2$. Independent errors add their variances. A rectangular mesh has $N(N-1)$ internal/external phases plus N output phases $= N^2$ phase shifters, so

$$\boxed{1 - F \approx (N-1)\,\sigma_\phi^2.}$$

3.2 Coupler errors

$B(\frac{1}{2}+\epsilon) = B(\frac{1}{2})\, e^{-i\epsilon\sigma_x'}$ (a small extra rotation with eigenvalues ± 1), so $\mathrm{Tr}W = N - 2 + 2\cos\epsilon$, giving $2\epsilon^2/N$ per coupler. With $N(N-1)$ couplers:

$$\boxed{1 - F \approx 2(N-1)\,\sigma_\eta^2.}$$

Check (Fig. 2b,c): the simulated ratio to these formulas at small σ is 0.99–1.01 for $N = 4, 6, 8$, for both error types. They stay accurate up to $1 - F \approx 0.1$.

3.3 What these numbers mean

- Infidelity grows like σ^2, but the **intensity error grows like** σ: for $N = 8$, $\sigma_\phi = 0.0126$ rad gives $1-F = 0.001$ but 1.05% of the light in wrong ports; $\sigma_\phi = 0.045$ gives 1.4% infidelity and 3.7% misrouted light. [Simulated]
- To keep $F > 0.99$ at $N = 8$ you need $\sigma_\phi \lesssim 0.038$ rad per phase shifter and $\sigma_\eta \lesssim 0.027$ per coupler. [Derived from the formulas]
- **Compare with Day 6:** an array-based arm 10 coupling lengths long with on-site errors $\sigma_\beta = 0.01\kappa_2$ already has an arm-to-arm error of ≈ 0.17 rad — more than four times this budget. Long arrays inside interferometer arms are therefore only sensible if their phase is protected, or if they are short. Days 9–10 test this properly.

4 Hardware: real MZI meshes

Not ours. The photos shown in our chat come from published work:

- The **programmable nanophotonic processor** of Shen *et al.*, *Nature Photonics* **11**, 441 (2017) — a silicon chip of MZIs used as an optical neural network layer; its accuracy was limited by phase-setting errors (your reference [2]).

- **Thermo-optic phase shifters**: metal heaters on top of silicon waveguides, wire-bonded to electronics; each heater sets one θ or ϕ.

In such chips, the dominant errors are exactly the two we modelled: phase errors (heater calibration, thermal crosstalk, width errors in arms) and coupler errors (gap errors in the directional couplers). Cite the source papers shown with each image before reusing a photo.

5 Code and tests

New: `topq/mzi.py` (coupler, phase, MZI, embedding, Clements layout, mesh, random settings, fidelity, intensity error, unitarity error); `scripts/day8_mzi_mesh.py` (110 s). `tests/test_mzi.py`: 23 tests, including the coupled-mode origin of the coupler, the closed form, the $4\eta(1-\eta)$ extinction limit, and both first-order fidelity laws. Suite: **119 passed**.

6 Self-test

1. Derive B from the two-guide coupled-mode equation. What $\kappa\ell$ gives 50:50?
2. Why can a gap error be written as a splitting error $\sigma_\eta = \frac{\pi}{4}\sigma_\kappa$?
3. Show that $BP(\theta)B$ has bar power $\sin^2(\theta/2)$.
4. Why is the bar state still perfect with equal imperfect couplers, but the cross state is not?
5. Why does $1 - F$ grow like σ^2 while the intensity error grows like σ?
6. For $N = 8$, what phase error per shifter keeps $F > 0.99$?